Learning with Unknown Input Observers for Robust Nonlinear Estimation

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Abstract: This paper proposes a hybrid estimator for nonlinear systems affected by unknown dynamics. A regularized unknown input observer (UIO) first provides physics-consistent estimates of the state and unknown input, even when the standard rank condition is not directly satisfied. These estimates then supervise an online neural correction, yielding a teacher–student architecture that combines robustness and adaptability. The method is specialized to vehicle state estimation by separating tire-force residuals from remaining modeling errors. Compact LMI conditions certify the observer gains, while the vehicle case study shows how the UIO-generated labels are used for residual learning.

Keywords: Robust estimation; Unknown input observer; Model learning; Neural network; Vehicle dynamics.

1. INTRODUCTION

Reliable vehicle state estimation is difficult when the available sensors do not reveal all the forces acting on the vehicle. Tire-road friction changes, load transfer, biases, delays, and discretization errors can all enter the dynamics as unknown inputs. Physics-based observers give useful robustness guarantees, but classical unknown input observers require structural rank conditions that are often too restrictive for practical sensing configurations (Hou and Müller, 1992; Trinh et al., 2008; Phanomchoeng and Rajamani, 2014). Data-driven estimators can learn nonlinear mismatch, yet pure learning approaches provide limited guarantees during online adaptation (Abdollahi et al., 2006; Buisson-Fenet et al., 2021).

This paper combines both viewpoints. The first layer is a generalized UIO that reconstructs the state and an unknown input estimate. When the standard condition $\mathrm{rank}(CB) = \mathrm{rank}(B)$ is not available, the model is regularized so that the residual effect is explicitly bounded. The second layer is a neural observer trained by the UIO estimate rather than by an unavailable ground truth disturbance. The chapter result in `ch4_main_copy.tex` is merged here as a compact vehicle case study: the UIO teacher estimates tire-only residual forces, while the neural student learns the remaining mismatch from teacher-generated correction labels.

The contribution is threefold: (i) a regularized continuous-time UIO design with an $\mathcal{H}^1$ error bound; (ii) a UIO-supervised neural residual learner with an $\mathcal{L}_2$ observer design condition; and (iii) a vehicle implementation that separates tire residual reconstruction from data-driven correction.

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2. REGULARIZED UIO DESIGN

Consider the nonlinear system

$$\dot{x} = \varphi(x,u) + B\mu_x, \qquad y = Cx, \tag{1}$$

where $x \in \mathbb{R}^{n_x}$, $u \in \mathbb{R}^{n_u}$, $y \in \mathbb{R}^{n_y}$, and $\mu_x \in \mathbb{R}^{n_\mu}$ is an unknown nonlinear term collecting unmodeled dynamics and disturbances. The matrices B and C are known. We assume that $n_\mu \leq n_y$, that $\varphi(\cdot,u)$ is Lipschitz on the operating set, and that μ_x is $\mathcal{L}_2$-bounded.

Direct unknown-input reconstruction usually requires

$$\mathrm{rank}(CB) = \mathrm{rank}(B). \tag{2}$$

To avoid losing feasibility when (2) fails, we use the regularized descriptor representation

$$\begin{aligned} \dot{x} + E_{\delta_1}\dot{\mu}_x &= \varphi(x,u) + B\mu_x + E\omega_{\delta_1}, \\ y &= Cx + D_{\delta_2}\mu_x + D\omega_{\delta_2}, \end{aligned} \tag{3}$$

where $E_{\delta_1} = \delta_1 E$, $D_{\delta_2} = \delta_2 D$, $\omega_{\delta_1} = \delta_1\dot{\mu}_x$, and $\omega_{\delta_2} = \delta_2\mu_x$. The matrices E and D are selected so that, with

$$\begin{gathered} \zeta = \begin{bmatrix} x \\ \mu_x \end{bmatrix}, \quad \mathbb{E} = [I_{n_x}\ E_{\delta_1}], \quad \mathcal{H} = [C\ D_{\delta_2}], \\ \bar{E} = [E\ 0], \quad \bar{D} = [0\ D], \quad \boldsymbol{\omega}_\delta = \begin{bmatrix} \omega_{\delta_1} \\ \omega_{\delta_2} \end{bmatrix}, \end{gathered} \tag{4}$$

the rank condition $\mathrm{rank}([\mathbb{E}^\top\ \mathcal{H}^\top]^\top) = n_x + n_\mu$ holds. Then (3) becomes

$$\mathbb{E}\dot{\zeta} = \varphi_\zeta(\zeta,u) + \bar{E}\boldsymbol{\omega}_\delta, \qquad y = \mathcal{H}\zeta + \bar{D}\boldsymbol{\omega}_\delta, \tag{5}$$

with $\varphi_\zeta(\zeta,u) = \varphi(x,u) + B\mu_x$.

Let P_ζ and Q_ζ satisfy

$$P_\zeta\mathbb{E} + Q_\zeta\mathcal{H} = I_{n_x+n_\mu}. \tag{6}$$

The observer is

$$\dot{\eta} = P_\zeta\varphi_\zeta(\hat{\zeta},u) + K(y - \mathcal{H}\hat{\zeta}), \qquad \hat{\zeta} = \eta + Q_\zeta y. \tag{7}$$

Defining $\tilde{\zeta} = \zeta - \hat{\zeta}$ and $\xi = \tilde{\zeta} + Q_\zeta\bar{D}\boldsymbol{\omega}_\delta$, the error dynamics can be written as

$$\dot{\xi} = (P_\zeta A_j - K\mathcal{H})\xi + (\bar{E} - P_\zeta A_j Q_\zeta \bar{D} - K\mathcal{D})\boldsymbol{\omega}_\delta \tag{8}$$

for each vertex A_j of a polytopic enclosure of the Jacobian of φ_ζ, where $\mathcal{D} = (I + \mathcal{H}Q_\zeta)\bar{D}$.

***Theorem** 1.* If there exist $\mathcal{P} = \mathcal{P}^\top > 0$, $\mathcal{Q}$, $\lambda > 0$, and a fixed $\epsilon > 0$ such that, for all Jacobian vertices A_j,

$$\begin{bmatrix} \Phi_j & \mathcal{P}\mathcal{E}_j - \mathcal{Q}^\top \mathcal{D} & \Psi_j \\ \star & -\lambda I & \mathcal{E}_j^\top \mathcal{P} - \mathcal{D}^\top \mathcal{Q} \\ \star & \star & -\epsilon^{-1}\mathcal{P} \end{bmatrix} < 0, \tag{9}$$

where

$$\Phi_j = (P_\zeta A_j)^\top \mathcal{P} + \mathcal{P}(P_\zeta A_j) - \mathcal{H}^\top \mathcal{Q} - \mathcal{Q}^\top \mathcal{H},$$

$$\Psi_j = (P_\zeta A_j)^\top \mathcal{P} - \mathcal{H}^\top \mathcal{Q}, \qquad \mathcal{E}_j = \bar{E} - P_\zeta A_j Q_\zeta \bar{D},$$

then the observer gain $K = \mathcal{P}^{-1}\mathcal{Q}^\top$ ensures

$$\|\tilde{\zeta}\|_{\mathcal{H}^1} \leq \lambda_\delta \|\mu_x\|_{\mathcal{H}^1} + \beta\|\xi(0)\|. \tag{10}$$

The constants λ_δ and β are proportional to $\max(\delta_1, \delta_2)$ and $\sqrt{\lambda_{\max}(\mathcal{P})}$, respectively.

Proof. With $V = \xi^\top \mathcal{P}\xi$ and $\mathcal{S} = \epsilon\mathcal{P}$, the Schur complement of (9) gives

$$\dot{V} + \|\xi\|^2 + \dot{\xi}^\top \mathcal{S}\dot{\xi} - \lambda\|\boldsymbol{\omega}_\delta\|^2 \leq 0.$$

Integration on $[0, \infty)$ yields an $\mathcal{H}^1$ bound for ξ. Since $\tilde{\zeta} = \xi - Q_\zeta \bar{D}\boldsymbol{\omega}_\delta$ and $\|\boldsymbol{\omega}_\delta\|_{\mathcal{L}_2} \leq \max(\delta_1, \delta_2)\|\mu_x\|_{\mathcal{H}^1}$, (10) follows.

3. GENERALIZED OUTPUT-DERIVATIVE UIO

The previous regularization is useful when (2) is not satisfied, but it may not give zero-error convergence. A less conservative construction augments the system with $\chi_1 = y$ and $\chi_2 = \dot{y}$:

$$\begin{aligned} \dot{\chi}_1 &= \chi_2, \\ \dot{\chi}_2 &= C\bar{\varphi}_x(x,u) + C\frac{\partial \varphi}{\partial x}(x,u)B\mu_x + CB\dot{\mu}_x, \\ \dot{x} &= \varphi(x,u) + B\mu_x, \\ y_\chi &= \chi_1 + C_\chi \chi_2 + Cx, \end{aligned} \tag{11}$$

where $\bar{\varphi}_x = (\partial\varphi/\partial x)\varphi$. Applying the same regularization to the $\dot{\chi}_2$ equation gives another descriptor model of the form (5), now with

$$\zeta = [\chi_1 \ \chi_2 \ x \ \mu_x]^\top.$$

The observer (7) and LMI (9) apply directly after replacing the corresponding matrices. If $\operatorname{rank}(CB) = \operatorname{rank}(B)$, one can set $\delta_1 = \delta_2 = 0$ with a suitable C_χ, so $\boldsymbol{\omega}_\delta \equiv 0$ and the error converges exponentially. If the rank condition fails, δ_1 and δ_2 tune the residual error bound. The auxiliary state $\hat{\chi}_2$ also provides a filtered estimate of $\dot{y}$, avoiding direct numerical differentiation of the measured output.

4. UIO-SUPERVISED RESIDUAL LEARNING

The UIO estimate is not the final model; it is a robust teacher for learning the repeatable part of μ_x. Decompose the generalized UIO output as

$$\hat{\zeta} = [\hat{\chi}_1 \ \hat{\chi}_2 \ \hat{x}_{\text{UIO}} \ \hat{\mu}_x]^\top. \tag{12}$$

A neural model $\mu_\theta = \mathcal{N}_\theta(\hat{x}_{\text{UIO}}, \hat{\mu}_x, u)$ is inserted in a student observer

$$\dot{\hat{x}}_{\text{NN}} = \varphi(\hat{x}_{\text{NN}}, u) + B\mu_\theta + L_{\text{NN}}(y - C\hat{x}_{\text{NN}}). \tag{13}$$

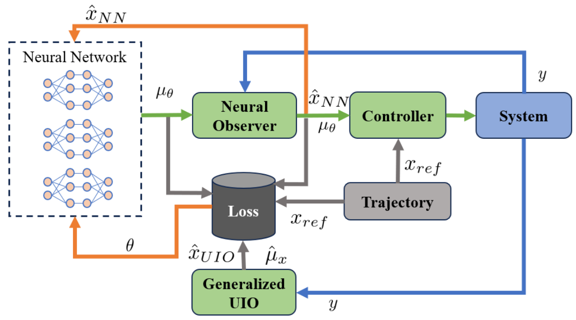

Fig. 1. Hybrid UIO-supervised learning architecture.

Writing the approximation error as $\Delta_x = \mu_x - \mu_\theta$, the student error $\epsilon_{\text{NN}} = x - \hat{x}_{\text{NN}}$ satisfies

$$\dot{\epsilon}_{\text{NN}} = (A_j^x - L_{\text{NN}}C)\epsilon_{\text{NN}} + B\Delta_x \tag{14}$$

for each vertex A_j^x enclosing the Jacobian of φ.

***Theorem** 2.* If there exist $\mathcal{P} = \mathcal{P}^\top > 0$, $\mathcal{Q}$, and $\sigma > 0$ such that, for all j,

$$\begin{bmatrix} (A_j^x)^\top \mathcal{P} + \mathcal{P}A_j^x - C^\top \mathcal{Q} - \mathcal{Q}^\top C & \mathcal{P}B \\ B^\top \mathcal{P} & -\sigma I \end{bmatrix} < 0, \tag{15}$$

then $L_{\text{NN}} = \mathcal{P}^{-1}\mathcal{Q}^\top$ gives

$$\|\epsilon_{\text{NN}}\|_{\mathcal{L}_2} \leq \sqrt{\sigma}\|\Delta_x\|_{\mathcal{L}_2} + \sqrt{\lambda_{\max}(\mathcal{P})}\|\epsilon_{\text{NN}}(0)\|.$$

The network parameters are updated with a composite loss

$$\begin{aligned} \mathcal{L}(\theta) =& \|y - C\hat{x}_{\text{NN}}\|_{T_y}^2 + \|\hat{x}_{\text{NN}} - \hat{x}_{\text{UIO}}\|_{T_u}^2 \\ &+ \lambda_\mu\|\mu_\theta - \hat{\mu}_x\|^2 + \lambda_\theta\|\theta\|^2, \end{aligned} \tag{16}$$

where the UIO consistency and residual terms prevent the student from drifting while the online learner adapts.

5. VEHICLE TEACHER–STUDENT CASE STUDY

The vehicle result is summarized with the state, input, and measurement vectors

$$\begin{aligned} x &= [v_x \ v_y \ r \ \psi \ X \ Y]^\top, \\ u &= [a \ \delta]^\top, \qquad y = [v_x \ r \ \psi \ X \ Y]^\top. \end{aligned} \tag{17}$$

The discrete vehicle model is written as

$$x_{k+1} = f_{\text{nom}}(x_k, u_k) + B_{\text{tire}}(x_k, u_k)w_k + G(x_k, u_k)d_k + \nu_k, \tag{18}$$

where $w_k = [\Delta F_{yf,k} \ \Delta F_{yr,k}]^\top$ contains tire-only residual forces and d_k collects the remaining mismatch. In continuous time, the tire residual channel is induced by

$$\begin{bmatrix} \dot{v}_x \\ \dot{v}_y \\ \dot{r} \end{bmatrix} = f_{\text{nom,dyn}} + \begin{bmatrix} 0 & 0 \\ 1/m & 1/m \\ l_f/I_z & -l_r/I_z \end{bmatrix} \begin{bmatrix} \Delta F_{yf} \\ \Delta F_{yr} \end{bmatrix}. \tag{19}$$

Thus the teacher residual is physically interpretable and cannot become a generic error bucket.

Layer 1 is an augmented-state EKF/UIO teacher. With $F_k = f_{\text{nom}}(x_k, u_k) + B_{\text{tire}}(x_k, u_k)w_k$,

$$\chi_k = [x_k^\top \ w_k^\top]^\top, \qquad \chi_{k+1} = [F_k^\top \ w_k^\top]^\top + q_k. \tag{20}$$

The teacher is not influenced by the neural model. It gives $\hat{x}_{k|k}^{(1)}$ and $\hat{w}_{k|k}$ and forms the label

$$r_k = S\left(\hat{x}_{k+1|k+1}^{(1)} - \hat{x}_{k+1|k}^{(1)}\right), \tag{21}$$

where S selects (v_x, v_y, r). This label captures the correction left after nominal dynamics and tire residual reconstruction.

Layer 2 is the student estimator

$$\hat{x}^{(2)}_{k+1|k} = f_{\mathrm{nom}}(\hat{x}^{(2)}_{k|k}, u_k) B_{\mathrm{tire}}(\hat{x}^{(2)}_{k|k}, u_k) \hat{w}_{k|k} S^\top r_\theta(z_k), \quad (22)$$

where $z_k = [\hat{x}^{(1)}_{k|k}\ u_k\ \rho_k\ \mathrm{flags}_k]^\top$. The online regression

$$\min_\theta \sum_k \alpha_k \|r_\theta(z_k) - r_k\|^2 \quad (23)$$

uses gates $\alpha_k \in [0, 1]$ to reject low-speed samples, yaw wrapping events, or large GNSS/IMU innovations.

In the simulation study from the merged chapter, the baseline physics-only observer tracks measured channels but shows larger lateral-state deviations under tire and parameter mismatch. The two-layer student reduces these deviations by injecting r_θ, while the teacher keeps the tire-force reconstruction independent of the learned correction. Residual accumulation in the learning update smooths transient spikes, and the loss decreases during online training, confirming that the UIO-generated labels provide a usable supervision signal for the remaining mismatch.

6. CONCLUSION

This paper condensed a regularized UIO, an output-derivative generalized UIO, and a UIO-supervised neural residual learner into a single IFAC-length framework. The merged vehicle case study separates tire-only residual estimation from neural learning of leftover mismatch. This preserves a stable physics-based teacher while allowing the student observer to improve prediction online. Future work will validate the method on real driving data and convert the continuous-time guarantees into a discrete-time implementation suitable for embedded deployment.

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